TITLE PAGE

- **Problem Statement ID – Question 4A**
- **Problem Statement Title- AI-Powered Behavioral Anomaly Detection for Cybersecurity**
- **Theme- Cybersecurity / Artificial Intelligence & Machine Learning**
- **PS Category- Software**
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- **Student ID- 23BCY10006**

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

- SENTINEL — Explainable Behavioral Anomaly Detection for Cybersecurity

- **Detailed explanation:** a behavioral profiler builds a rolling per-entity "normal" profile (login hours, geo, resources, device), a dual detector (GRU-Autoencoder + Isolation Forest) scores deviations unsupervised, a LightGBM classifier names the attack type, and SHAP explains every alert in plain English on a live SOC dashboard.

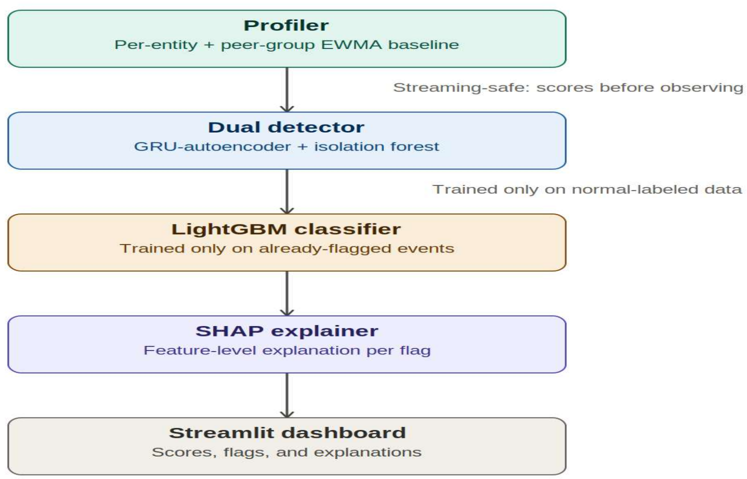
- **How it addresses the problem:** detects credential misuse, brute force, impossible travel, lateral movement, and device spoofing in near real-time, without ever training on the imbalanced attack labels directly.

- **Innovation:** cold-start entities blend into a peer-group baseline instead of going unscored, profiles decay/adapt to legitimate drift, and every alert ships with a SHAP-based plain-English reason, not just a bare score.

TECHNICAL APPROACH

- **Technologies:** Python, pandas/NumPy for feature engineering; PyTorch GRU-Autoencoder (sequence-aware detector) + scikit-learn Isolation Forest (fast baseline); LightGBM for attack-type classification; SHAP (TreeExplainer) for explainability; Streamlit + Plotly for the live analyst dashboard.

- **Methodology:**



FEASIBILITY AND VIABILITY

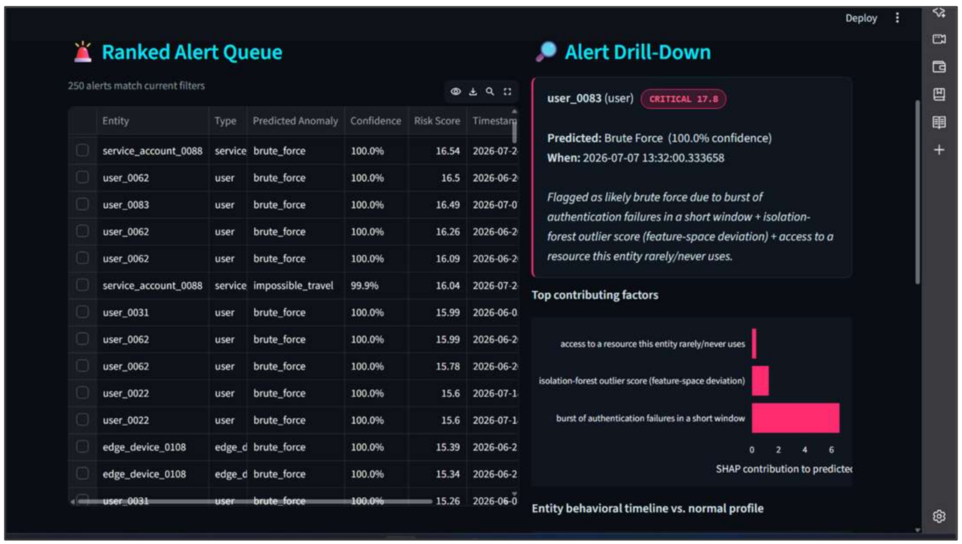
- **Feasibility:** prototype is built and running end-to-end on synthetic data (11.7K events, 175 entities) — ROC-AUC 0.99, precision 0.82, recall 0.87 at a realistic ~5% alert budget, 76% attack-type classification accuracy. All components use mature, production-proven libraries (PyTorch, scikit-learn, LightGBM, SHAP, Streamlit).

- **Challenges & risks:** synthetic-to-real gap (attack "mechanics" are hand-designed, e.g. fixed impossible-travel speed threshold); our honest ablation shows Isolation Forest currently edges out the GRU-Autoencoder on PR-AUC (0.685 vs 0.483) at this data scale — most of our 7 attack types are single-event outliers, not yet deeply sequential; cold-start entities have a narrow but real under-scored window.

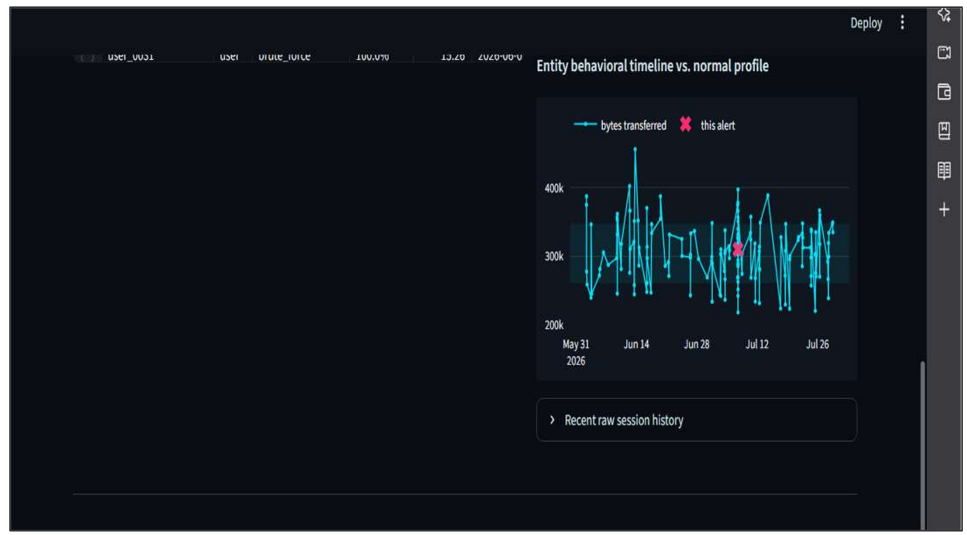
- **Strategies for overcoming these challenges:** validate on red-team/live traffic before production; use IF as the low-latency first-pass filter with GRU-AE as a secondary/tie-breaker signal, growing its weight as per-entity history deepens; ramp cold-start entities from peer-group to personal baseline over their first 15 sessions.

ARTIFACTS

- **Code embedded:** data_generator.py, profiler.py, detector.py, classifier.py, explain.py, evaluate.py, dashboard.py (submitted alongside this deck)
- **Live SOC dashboard snaps below** — real screenshots of the running Streamlit console, not mockups



Ranked alert queue + SHAP-based explainability drill-down (live console)



Entity behavioral timeline vs. normal profile (sequence view)

RESEARCH AND REFERENCES

- Liu, Ting & Zhou (2008), "Isolation Forest", IEEE ICDM — basis for our fast baseline anomaly detector.
- Malhotra et al. (2016), "LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection" — informed our sequence-aware GRU-Autoencoder design.
- Lundberg & Lee (2017), "A Unified Approach to Interpreting Model Predictions" (SHAP), NeurIPS — powers our explainability layer.
- MITRE ATT&CK framework — reference taxonomy for mapping detected behaviors (credential access, lateral movement) to known adversary techniques.
- Gartner's User and Entity Behavior Analytics (UEBA) category — industry framing for behavioral, baseline-driven detection over signature rules.